



Alternate periodic event-triggered control for synchronization of multilayer neural networks

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ABSTRACT

In this paper, the exponential synchronization of multilayer neural networks (MNNs) is investigated via alternate periodic event-triggered control (APETC). Distinguished from the previous work, a novel APETC which incorporates aperiodically intermittent control (AIC) and periodic event-triggered mechanism is firstly proposed. Determined by two event-triggered conditions, the control and rest intervals of APETC are based on the present state of the system rather than being predetermined. In contrast to the conventional event-triggered control (ETC), the event-triggered conditions of APETC can not only judge the updates of control signals, but also dominate the actuation and close of the controller. Moreover, by introducing the sampling period into ETC, the number of event triggers can be decreased and the Zeno phenomenon is completely avoided. Subsequently, synchronization criteria for MNNs under APETC are established on the basis of Lyapunov method and graph theory. Finally, several numerical simulations are performed to demonstrate the validity of the theoretical results.

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1. Introduction

Neural networks have attracted noticeable attention over the past few years for their broad applications in the fields of biology, chemistry, architecture, artificial intelligence and so on [1,14,18,24,29]. Recently, numerous remarkable results in relation to single-layer neural networks have been received [1,14,18]. In fact, sets of networks with diverse kinds of connections in the real world tend to be interrelated with each other, which can not be simply confined into single-layer networks. For instance, in a metropolis, the bus and the underground systems which possess different routes may have several identical stations, thus making for multilayer transportation networks. Moreover, [24] indicated that the biological neural networks in the cortex is multilayered, which illustrated the multilayer structures existing in neural networks are reasonable. In [30], multilayer networks were proposed to reveal previously hidden features of brain networks. A three-layer neuronal network was considered in [28] to investigate different patterns of communication between synapses. Consequently, MNNs have posed a series of issues that deserve further investigation in the areas of network and engineering science including cascading failures, spreading processes, synchronization and so on [5,26,27]. As a crucial dynamical behavior of neural networks, synchronization has been considerably studied by several scholars [31,39,42]. For instance, as seen in [39], synchronization of multilayer memristive neural networks was explored. However, owing to the complexity of topological structures

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of MNNs and inherent relations between layers, it is normally difficult for MNNs to reach self-synchronization. Accordingly, it is essential to design an appropriate and valid control strategy to realize the synchronization of MNNs.

Up to now, multifarious control strategies have been employed to reach the desired dynamical behaviors, such as optimal control [6], impulsive control [44], pinning control [32], intermittent control [21] and so on. Intermittent control (IC), where the control is only implemented during a sequence of disjoint time intervals which are non-zero, constitutes an effective discontinuous control strategy to reach synchronization favored by many researchers [9,21,45]. In contrast to the continuous control strategies, discontinuous control strategies can reduce the information transmissions and increase the economic benefits, which make them more sought-after in applications. Additionally, IC is a better alternative in secure communication due to its increased challenges in capturing transmitted signals in the process of transferring data. Generally, IC can be categorized into periodically intermittent control (PIC) [8,21] and aperiodically intermittent control (AIC) [33,47]. For example, based on PIC, exponential synchronization was considered in [22]. Moreover, PIC was also used to investigate the stabilization of a class of highly nonlinear systems [21]. However, it should be pointed out that the limitations of PIC on the fixed control period might be inappropriate in some practical issues. A typical instance is represented by the generation of solar energy affected by the external perturbation of the real world unavoidably, which is evidently aperiodically intermittent [2]. Besides, PIC is subordinated to AIC in a sense. Consequently, from both practical and theoretical perspectives, it is meaningful to research synchronization of MNNs via AIC.

However, the constraints on the lower bound of control widths and upper bound of control periods or the maximum proportion of rest widths are indispensable assumptions in existing research efforts on AIC. To conceptualize, the mathematical descriptions are given as follows:

1. $\inf_i(s_i - t_i) = \vartheta, \sup_i(t_{i+1} - t_i) = \varpi$, where $0 < \vartheta < \varpi < +\infty$ [19,20],
2. $\limsup_{i \rightarrow +\infty} \frac{t_{i+1} - s_i}{t_{i+1} - t_i} = \Omega \in (0, 1)$ [4,34],

where $(t_i, s_i]$ is the i -th control interval and $(s_i, t_{i+1}]$ is the i -th rest interval. In fact, the assumptions which involve limits operator require complicated computations to verify and lack complete aperiodicity as the proactive control and rest intervals are inflexible. Also, provided that the sequence of time instants is determined according to the assumptions, the designed AIC will be operated following with the sequence, which might be conservative and unnecessary in practice. Apart from the restraints of the assumptions, the traditional AIC based on continuous-time state during control intervals may result in needless energy consumption, which contributes to the development of aperiodically intermittent time-triggered control (AITTC) when investigating the synchronization problem. For example, in [38], AITTC was studied for stochastic signed networks and in [7], Chen et al. investigated AITTC for heterogeneous complex networks. However, the sampling time interval of AITTC is generically a fixed constant during control intervals, which might lead to limitations in aspect of applications.

To avoid the constraints of AIC mentioned above, event-triggered control (ETC), which can better coincide with reality since it generates times for the updates of control signals based on the present state of the plant rather than sampled time instants, has been introduced into AIC [11,36]. ETC can not only save the utilization of communication resources compared to the traditional time-triggered control, but also reduce the limitations of AITTC with desirable performance of the closed-loop system. Therefore, ETC has drawn incremental attention in the last few years [3,12,13,25,43]. In addition, the studies on the stability and synchronization of neural networks were given in [16,35,46]. It is worth noting that in the majority of the existing literature, ETC can be termed continuous event-triggered control (CETC) as the event-triggered condition needs to be evaluated continuously. For CETC, it is essential to guarantee a positive minimum inter-event time by multiple methods [17,23,43] to exclude infinite fast sampling such as the Zeno phenomenon. Recently, sampled-data-based ETC (also called periodic event-triggered control) has been widely investigated. In comparison with CETC, periodic event-triggered control (PETC), whose event-triggered condition is validated periodically and new state measurements, control signals are decided whether to transmit or not at each sampling time, can avoid the continuous monitoring on networks. It should be pointed out that the sampling period in PETC ensures the existence of a positive minimum inter-event time between execution times, which can naturally avoid the Zeno phenomenon. Additionally, the number of event triggers of PETC is less than that under the corresponding CETC, which can further reduce the usages of communication resources. In view of the advantages of PETC above, introducing PETC into AIC is of significant value.

The objective of this paper is to propose a novel alternate periodic event-triggered control (APETC) which combines the advantages of AIC and PETC. For the designed APETC mechanism, it consists of two alternate event-triggered conditions. While the AIC introducing a single event-triggered condition can merely decide the updates of control signals, APETC can also drive the controller to be in the working state when the control event-triggered condition is satisfied and off when it meets the rest event-triggered condition. Therefore, the control and rest intervals depend on the state and performance of the system instead of being predesigned. In addition, APETC can alleviate the constraints that the assumptions of AIC bring about. In comparison with the AIC based on time-triggered control, where the system state is sampled each regular time interval during control intervals, APETC can mitigate the unnecessary updates of control signals and the waste of resources. Furthermore, APETC is free from the Zeno phenomenon as it derives a guaranteed positive minimum inter-event time of the sampling interval directly. However, to the best of our knowledge, there is no result in regard to the synchronization of MNNs by employing APETC.

Motivated by the analysis above, a novel APETC for investigating the synchronization of MNNs is firstly proposed in this paper. Based on Lyapunov method and graph theory, several sufficient conditions for the synchronization of MNNs are obtained. The main contributions and innovations of this paper can be stated into four points:

- For the topology of MNNs, it is only required that the composite digraph of subnetworks is strongly connected, while the subdigraphs are not constrained to be strongly connected, which decreases the limitations on the topology of subnetworks.
- A novel APETC which removes the limitations of IC [34,40] is firstly designed. Unlike the fixed or predetermined control sequence in intermittent control, the control sequence of APETC is generated by event-triggered mechanism. The two event-triggered conditions of APETC can not only dominate the updates of control signals, but also determine the actuation and close of the controller, thus greatly lowering the energy consumption and communication burden.
- While the conventional CETC [12,16] needs multiple methods to exclude the Zeno phenomenon, the PETC we adopt can eliminate the Zeno phenomenon in essence by deriving the minimum inter-execution time directly, namely $t_{k+1} - t_k \geq \tau_1$. Subsequently, two estimations of the upper bounds of sampling error and the lower bounds of state error can be obtained.
- The parameter τ_1 has a significant influence on the exponential synchronization of MNNs. The upper bound of the minimum inter-execution time τ_1^* is determined by the unique solution to the equation

$$-(\lambda - \gamma_1)G_2(\tau_1) + \frac{1 + \varepsilon}{\varepsilon}dG_1(\tau_1) + \frac{2d\delta_1(1 + \varepsilon)}{1 - 2\delta_1}$$

where the parameters can be found in the main results.

The remainder of the paper is arranged as follows: in Section 2, some essential preliminaries and model formulation are given. Section 3 and Section 4 propose the main results of this paper. Then, some relevant numerical simulations are presented in Section 5 to illustrate the validity of the theoretical results. Finally, the conclusions of this paper are given in Section 6.

2. Preliminaries and model formulation

2.1. Notations

Throughout this paper, the following notations will be used. Denote $\mathbb{R} = (-\infty, +\infty)$, $\mathbb{R}^+ = [0, +\infty)$, $\mathbb{Z} = \{0, 1, 2, \dots\}$, $\mathcal{M} = \{1, 2, \dots, M\}$, $\mathcal{N} = \{1, 2, \dots, N\}$. Let $\mathbb{R}^n$ represent the n -dimensional Euclidean space and $\mathbb{R}^{m \times n}$ be the set of $m \times n$ real matrices. For any vector $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$, $|x| = \sqrt{\sum_{i=1}^n x_i^2}$ is the Euclidean norm of x . Additionally, the superscript “T” stands for the transpose of a vector or a matrix. Denote $a \vee b$ to be the maximum of a and b . Let $\lceil c \rceil$ be the minimum integer greater than or equal to c and $\lfloor c \rfloor$ be the maximum integer less than or equal to c .

A digraph $\mathcal{G} = (\mathcal{N}, E)$ includes a set $\mathcal{N}$ of nodes and a set E of arcs (j, i) linking from node j to node i . A digraph $\mathcal{G}$ is weighted if each arc (j, i) is assigned a positive weight a_{ij} . Also, $a_{ij} > 0$ if and only if an arc (j, i) exists in digraph $\mathcal{G}$. For a weighted digraph $\mathcal{G}$ with N nodes, define the weight matrix $A = (a_{ij})_{N \times N}$, in which a_{ij} is equal to the weight of arc (j, i) , and $a_{ij} = 0$ if arc (j, i) is nonexistent. The digraph $\mathcal{G}$ with a weight matrix A can be denoted as $(\mathcal{G}, A)$ and the digraph $\mathcal{G}$ with several weight matrices $A_l, l \in \mathcal{M}$ is defined as $(\mathcal{G}, A_1, A_2, \dots, A_M)$, which can be separated into several subdigraphs $(\mathcal{G}, A_1), (\mathcal{G}, A_2), \dots, (\mathcal{G}, A_M)$. For any pair of different nodes of digraph $\mathcal{G}$, if there exists a directed edge from one to the other, the digraph $\mathcal{G}$ is strongly connected. More details in regard to graph theory can be found in [15].

2.2. Model formulation

Now, we consider the following MNNs

$$\dot{x}_i(t) = -b_i x_i(t) + f_i(x_i(t), t) + \sum_{l=1}^M r_l \sum_{j=1}^N a_{ij}^{(l)} x_j(t), i \in \mathcal{N}, \quad (1)$$

where $x_i(t) \in \mathbb{R}^n$ is the state vector of the i -th node, b_i represents the self feedback connection strength, $f_i: \mathbb{R}^n \times \mathbb{R}^+ \rightarrow \mathbb{R}^n$ denotes the activation function of the i -th node, $r_l > 0$ signifies the coupling strength, and a_{ij} is defined as follows: if there exists a directed edge from node j to node i in the l -th layer, then $a_{ij}^{(l)} > 0$; otherwise, $a_{ij}^{(l)} = 0$.

System (1) is considered as the drive system, then the corresponding response system can be obtained as follows

$$\dot{y}_i(t) = -b_i y_i(t) + f_i(y_i(t), t) + \sum_{l=1}^M r_l \sum_{j=1}^N a_{ij}^{(l)} y_j(t) + u_i(t), i \in \mathcal{N}, \quad (2)$$

where $y_i(t) \in \mathbb{R}^n$ and $u_i(t)$ is a control input, which will be defined later.

Define the synchronization error as $\rho_i(t) = y_i(t) - x_i(t)$. Then, according to drive system (1) and response system (2), the error system can be described as

$$\dot{\rho}_i(t) = -b_i \rho_i(t) + f_i(y_i(t), t) - f_i(x_i(t), t) + \sum_{l=1}^M r_l \sum_{j=1}^N a_{ij}^{(l)} \rho_j(t) + u_i(t), i \in \mathcal{N}. \quad (3)$$

Next, the ETC $u_i(t)$ is designed as follows

$$u_i(t) = -d_i \rho_i(t_k), t \in [t_k, t_{k+1}), \quad (4)$$

where $d_i > 0$ signifies the control gain.

Then, the event-triggered instant t_k is determined by the sequence

$$t_{k+1} = \inf \left\{ m\tau_1 > t_k \mid \sum_{i=1}^N c_i |\rho_i(m\tau_1) - \rho_i(t_k)|^2 - \delta_1 \sum_{i=1}^N c_i |\rho_i(t_k)|^2 \geq 0 \right\}, \quad (5)$$

where $\delta_1 > 0$, τ_1 is a pre-specified positive constant, called the sampling period and c_i stands for the cofactor of the i -th diagonal element of the Laplacian matrix of the strongly connected digraph $(\mathcal{G}, A)$ with $A = (r_{ij})_{N \times N}$, $r_{ij} = \max\{a_{ij}^2, a_{ij}\}$, in which $a_{ij} = \max\{a_{ij}^{(1)}, \dots, a_{ij}^{(M)}\}$. Furthermore, as a matter of convenience, we set $e(t) = \rho(\Delta_t) - \rho(t)$, $\tilde{e} = \rho(\xi_t) - \rho(\Delta_t)$ and define Δ_t and ξ_t as the last sampling and execution instants before t respectively, that is, $\Delta_t = \max\{m\tau_1 \leq t, m \in \mathbb{Z}\}$ and $\xi_t = \max\{t_k \leq t, k \in \mathbb{Z}\}$. Before further analysis, an assumption and a definition are introduced.

Assumption 1. [41] For any $\mu, v \in \mathbb{R}^N, i = 1, 2, \dots, N$, there exists a positive constant ρ_{i1} such that

$$|f_i(\mu, t) - f_i(v, t)| \leq \rho_{i1} |\mu - v|.$$

Definition 1. [10] System (1) and System (2) are said to reach exponential synchronization if there exist two positive constants μ and λ such that

$$|\rho(t)|^2 \leq \mu e^{-\lambda t}, t \geq 0,$$

where λ represents the synchronization rate.

3. Periodic event-triggered control

In this section, based on graph theory and Lyapunov method, some sufficient conditions are provided to investigate the exponential synchronization of MNNs by PETC. Firstly, an important lemma is established before deriving our theoretical results.

Lemma 1. Assume that Assumption 1 holds. When $\sum_{i=1}^N c_i |\tilde{e}_i|^2 \neq 0$, for fixed τ_1 in the event-triggered mechanism (5) satisfying $t_{k+1} - t_k \geq \tau_1$, where $\tau \in [m\tau_1, (m+1)\tau_1]$ the following estimations can be obtained

$$\sum_{i=1}^N c_i |e_i(t)|^2 \leq G_1(\tau_1) \sum_{i=1}^N c_i |\rho_i(\Delta_t)|^2,$$

and

$$\sum_{i=1}^N c_i |\rho_i(t)|^2 \geq G_2(\tau_1) \sum_{i=1}^N c_i |\rho_i(\Delta_t)|^2,$$

where

$$\begin{aligned} G_1(\tau_1) &= 4\tau_1^2 \varepsilon_1 \exp\{\varepsilon_2 \tau_1\}, \\ G_2(\tau_1) &= e^{-\varepsilon_3 \tau_1} + \frac{2d}{\varepsilon_3(1-2\delta_1)} (e^{-\varepsilon_3 \tau_1} - 1), \end{aligned}$$

in which $\delta_1 \in (0, 0.5)$, $\varepsilon_1 = \max_{i \in \mathcal{N}^*} \{\varepsilon_{i1}\}$, $\varepsilon_{i1} = \frac{2d^2}{1-2\delta_1} + 2b_i^2 + 2\rho_{i1}^2 + 2M^2 N \Gamma^{*2} \sum_{j=1}^N r_{ij}$, $\varepsilon_2 = \max_{i \in \mathcal{N}^*} \{\varepsilon_{i2}\}$, $\varepsilon_{i2} = 8\tau_1(b_i^2 + \rho_{i1}^2 + \Gamma^{*2} M^2 N \sum_{j=1}^N r_{ij})$, $\varepsilon_3 = \max_{i \in \mathcal{N}^*} \{\varepsilon_{i3}\}$, $\varepsilon_{i3} = 2b_i + 2\rho_{i1} + d_i + 2\Gamma^* M \sum_{j=1}^N r_{ij}$, $d = \max_{i \in \mathcal{N}^*} \{d_i\}$, and $\Gamma^* = \max_{i \in \mathcal{N}^*} \{r_i\}$.

Proof. When $\sum_{i=1}^N c_i |\tilde{e}_i|^2 \neq 0$, we can obtain

$$\begin{aligned}\sum_{i=1}^N c_i |\rho_i(\xi_t)|^2 &\leq 2 \sum_{i=1}^N c_i (|\tilde{e}_i|^2 + |\rho_i(\Delta_t)|^2) \\ &\leq 2\delta_1 \sum_{i=1}^N c_i |\rho_i(\xi_t)|^2 + 2 \sum_{i=1}^N c_i |\rho_i(\Delta_t)|^2.\end{aligned}$$

Hence, this yields

$$\sum_{i=1}^N c_i |\rho_i(\xi_t)|^2 \leq \frac{2}{1-2\delta_1} \sum_{i=1}^N c_i |\rho_i(\Delta_t)|^2. \quad (6)$$

According to [Assumption 1](#), it can be deduced that

$$\begin{aligned}|e_i(t)|^2 &\leq \left| \int_{\Delta_t}^t \left[-b_i \rho_i(s) + f_i(y_i(s), s) - f_i(x_i(s), s) + \sum_{l=1}^M r_l \sum_{j=1}^N a_{ij}^{(l)} \rho_j(s) - d_i \rho_i(\xi_s) \right] ds \right|^2 \\ &\leq \tau_1 \int_{\Delta_t}^t \left(4b_i^2 |\rho_i(s)|^2 + 4|f_i(y_i(s), s) - f_i(x_i(s), s)|^2 + 4\Gamma^{*2} M^2 N \sum_{j=1}^N a_{ij}^2 |\rho_j(s)|^2 + 4d_i^2 |\rho_i(\xi_s)|^2 \right) ds \\ &\leq \int_{\Delta_t}^t \left(4\tau_1 (b_i^2 + \rho_{i1}^2) |\rho_i(s)|^2 + 4\tau_1 \Gamma^{*2} M^2 N \sum_{j=1}^N a_{ij}^2 |\rho_j(s)|^2 + 4\tau_1 d_i^2 |\rho_i(\xi_s)|^2 \right) ds \\ &\leq 4\tau_1^2 d_i^2 |\rho_i(\xi_t)|^2 + \int_{\Delta_t}^t 4\tau_1 \left(b_i^2 + \rho_{i1}^2 + \Gamma^{*2} M^2 N \sum_{j=1}^N r_{ij} \right) |\rho_i(s)|^2 ds \\ &\quad + 4\tau_1 \int_{\Delta_t}^t \Gamma^{*2} M^2 N \sum_{j=1}^N r_{ij} (|\rho_j(s)|^2 - |\rho_i(s)|^2) ds.\end{aligned} \quad (7)$$

Therefore, based on (7), we can derive

$$\begin{aligned}\sum_{i=1}^N c_i |e_i(t)|^2 &\leq 4\tau_1^2 \sum_{i=1}^N c_i d_i^2 |\rho_i(\xi_t)|^2 + \int_{\Delta_t}^t \sum_{i=1}^N c_i \left[4\tau_1 \left(b_i^2 + \rho_{i1}^2 + \Gamma^{*2} M^2 N \sum_{j=1}^N r_{ij} \right) \right] |\rho_i(s)|^2 ds \\ &\quad + 4\tau_1 \int_{\Delta_t}^t \Gamma^{*2} M^2 N \sum_{i=1}^N \sum_{j=1}^N c_i r_{ij} (|\rho_j(s)|^2 - |\rho_i(s)|^2) ds.\end{aligned}$$

According to Theorem 2.2 in [\[15\]](#), it is not difficult to yield that

$$\sum_{i=1}^N \sum_{j=1}^N c_i r_{ij} (|\rho_j(t)|^2 - |\rho_i(t)|^2) = 0. \quad (8)$$

By combining (6)–(8), it has

$$\begin{aligned}\sum_{i=1}^N c_i |e_i(t)|^2 &\leq 4\tau_1^2 \sum_{i=1}^N c_i d_i^2 |\rho_i(\xi_t)|^2 + \int_{\Delta_t}^t \sum_{i=1}^N c_i \left[4\tau_1 \left(b_i^2 + \rho_{i1}^2 + \Gamma^{*2} M^2 N \sum_{j=1}^N r_{ij} \right) \right] |\rho_i(s)|^2 ds \\ &\leq 4\tau_1^2 \varepsilon_1 \sum_{i=1}^N c_i |\rho_i(\Delta_t)|^2 + \int_{\Delta_t}^t \varepsilon_2 \sum_{i=1}^N c_i |e_i(s)|^2 ds.\end{aligned}$$

By employing Grönwall's inequality, it has

$$\sum_{i=1}^N c_i |e_i(t)|^2 \leq 4\tau_1^2 \varepsilon_1 \exp\{\varepsilon_2 \tau_1\} \sum_{i=1}^N c_i |\rho_i(\Delta_t)|^2.$$

On the other hand, set $V_i(t) = -|\rho_i(t)|^2$, and we can obtain

$$\begin{aligned}\dot{V}_i(t) &= -2\rho_i^T(t) \left(-b_i \rho_i(t) + f_i(y_i(t), t) - f_i(x_i(t), t) + \sum_{l=1}^M r_l \sum_{j=1}^N a_{ij}^{(l)} \rho_j(t) - d_i \rho_i(\xi_t) \right) \\ &\leq 2b_i |\rho_i(t)|^2 + 2\rho_{i1} |\rho_i(t)|^2 + d_i |\rho_i(t)|^2 + d_i |\rho_i(\xi_t)|^2 + \Gamma^* M \sum_{j=1}^N r_{ij} (|\rho_i(t)|^2 + |\rho_j(t)|^2) \\ &= (2b_i + 2\rho_{i1} + d_i + 2\Gamma^* M \sum_{j=1}^N r_{ij}) |\rho_i(t)|^2 + \Gamma^* M \sum_{j=1}^N r_{ij} (|\rho_j(t)|^2 - |\rho_i(t)|^2) + d_i |\rho_i(\xi_t)|^2.\end{aligned}$$

Therefore, it can be deduced that

$$\begin{aligned}\sum_{i=1}^N c_i \dot{V}_i(t) &\leq \varepsilon_3 \sum_{i=1}^N c_i |\rho_i(t)|^2 + \frac{2d}{1-2\delta_1} \sum_{i=1}^N c_i |\rho_i(\Delta_t)|^2 \\ &= -\varepsilon_3 \sum_{i=1}^N c_i V_i(t) + \frac{2d}{1-2\delta_1} \sum_{i=1}^N c_i |\rho_i(\Delta_t)|^2.\end{aligned}\quad (9)$$

Then, by applying the comparison principle to (9), we have

$$\sum_{i=1}^N c_i V_i(t) \leq \left[\frac{2d}{\varepsilon_3(1-2\delta_1)} (e^{-\varepsilon_3(t-\Delta_t)} - 1) + e^{-\varepsilon_3(t-\Delta_t)} \right] \sum_{i=1}^N c_i V_i(\Delta_t).$$

In other words, we can derive the following inequality

$$\sum_{i=1}^N c_i |\rho_i(t)|^2 \geq G_2(\tau_1) \sum_{i=1}^N c_i |\rho_i(\Delta_t)|^2.$$

The proof is completed.

Remark 1. Traditional researches on synchronization of neural networks are usually depicted by single-layer networks. However, MNNs include more complicated topology structures. Meanwhile, the introduction of multilayer interactions increases the difficulty of constructing the Lyapunov function. For MNNs, each subnetwork topology corresponding to each layer may be strongly connected or not. It is difficult to use graph theory directly when establishing the Lyapunov function of networks. In this paper, we consider the composite digraph of digraphs $(\mathcal{G}, (r_{ij}^{(1)})_{N \times N}), (\mathcal{G}, (r_{ij}^{(2)})_{N \times N}), \dots, (\mathcal{G}, (r_{ij}^{(M)})_{N \times N})$. It is only required that composite digraph $(\mathcal{G}, A)$ ($A = (r_{ij})_{N \times N}, r_{ij} = \max\{a_{ij}^2, a_{ij}\}, a_{ij} = \max\{a_{ij}^{(1)}, \dots, a_{ij}^{(M)}\}$) is strongly connected, while sub-digraphs are not required to be strongly connected. Based on the strongly connected digraph $(\mathcal{G}, A)$, we can construct the Lyapunov function successfully by using graph theory.

Remark 2. Recently, the dynamic properties of neural networks have been investigated in [16,35,46] by designing ETC. At present, ETC inevitably faces two specific problems. 1) How to estimate and dominate the sampling error? 2) How to avoid the Zeno phenomenon? Based on the above analysis, we consider PETC. In PETC, events are only generated after a predetermined time threshold has elapsed, which results in discrete sampling. This automatically guarantees the existence of a minimum inter-execution time and excludes the Zeno phenomenon naturally. In addition, two estimations about sampling error and state error are obtained in Lemma 1.

Next, the main results for the exponential synchronization of drive system (1) and response system (2) will be presented.

Theorem 1. Systems (1) and (2) can reach exponential synchronization under PETC mechanism (5), if Assumption 1 and the following conditions are satisfied

$$d_i + \psi_i > 0, \lambda > \gamma_1, \delta_1 \in \left(0, \frac{\lambda - \gamma_1}{2(d(1+\varepsilon) + (\lambda - \gamma_1))}\right),$$

in which γ_1 is a positive constant,

$$\lambda = \min_{i \in \mathcal{A}^*} \{d_i + \psi_i\}, \psi_i = 2b_i - 2\rho_{i1} - 2M\Gamma^* \sum_{j=1}^N r_{ij},$$

and $\tau_1 \in (0, \tau_1^*)$, where τ_1^* is the unique solution to the equation

$$-(\lambda - \gamma_1)G_2(\tau_1) + \frac{1+\varepsilon}{\varepsilon}dG_1(\tau_1) + \frac{2d\delta_1(1+\varepsilon)}{1-2\delta_1}$$

Proof. First of all, define Lyapunov function

$$V(t) = \sum_{i=1}^N c_i V_i(t),$$

where $V_i(t) = |\rho_i(t)|^2$.

Case I. When $\sum_{i=1}^N c_i |\tilde{e}_i|^2 = 0$, we can obtain

$$\begin{aligned} \dot{V}_i(t) &= 2\rho_i^T(t) \left(-b_i \rho_i(t) + f_i(y_i(t), t) - f_i(x_i(t), t) + \sum_{l=1}^M r_l \sum_{j=1}^N a_{ij}^{(l)} \rho_j(t) - d_i \rho_i(\xi_t) \right) \\ &\leq -2b_i |\rho_i(t)|^2 + 2|\rho_i(t)| |f_i(y_i(t), t) - f_i(x_i(t), t)| + 2\rho_i^T(t) \sum_{l=1}^M r_l \sum_{j=1}^N a_{ij}^{(l)} \rho_j(t) - 2d_i |\rho_i(t)|^2 + 2d_i |\rho_i(t)| |e_i(t)|. \end{aligned} \quad (10)$$

Given the [Assumption 1](#), the following inequalities can be derived

$$2|\rho_i(t)| |f_i(y_i(t), t) - f_i(x_i(t), t)| \leq 2\rho_{i1} |\rho_i(t)|^2 \quad (11)$$

and

$$\begin{aligned} &2\rho_i^T(t) \sum_{l=1}^M r_l \sum_{j=1}^N a_{ij}^{(l)} \rho_j(t) \\ &\leq M\Gamma^* \sum_{j=1}^N a_{ij} |\rho_i(t)|^2 + M\Gamma^* \sum_{j=1}^N a_{ij} |\rho_j(t)|^2 \\ &\leq 2M\Gamma^* \sum_{j=1}^N r_{ij} |\rho_i(t)|^2 + M\Gamma^* \sum_{j=1}^N r_{ij} (|\rho_j(t)|^2 - |\rho_i(t)|^2). \end{aligned} \quad (12)$$

By combining (10), (11) and (12), it has

$$\begin{aligned} \dot{V}_i(t) &\leq \sum_{i=1}^N c_i \left(-2b_i + 2\rho_{i1} + 2M\Gamma^* \sum_{j=1}^N r_{ij} - d_i \right) |\rho_i(t)|^2 \\ &\quad + d \sum_{i=1}^N c_i |e_i(t)|^2 + M\Gamma^* \sum_{i=1}^N \sum_{j=1}^N c_i r_{ij} (|\rho_j(t)|^2 - |\rho_i(t)|^2) \\ &\leq -\gamma_1 \sum_{i=1}^N c_i |\rho_i(t)|^2 - (\lambda - \gamma_1) G_2(\tau_1) \sum_{i=1}^N c_i |\rho_i(\xi_t)|^2 + dG_1(\tau_1) \sum_{i=1}^N c_i |\rho_i(\xi_t)|^2 \\ &\leq -\gamma_1 V(t), \end{aligned} \quad (13)$$

where in $G_1(\tau_1), \frac{2d^2}{1-2\delta_1} \rightarrow d^2$, and in $G_2(\tau_1), \frac{2d}{\varepsilon_3(1-2\delta_1)} \rightarrow \frac{d}{\varepsilon_3}$.

Case II. When $\sum_{i=1}^N c_i |\tilde{e}_i|^2 \neq 0$, similarly we can obtain

$$\begin{aligned} \dot{V}(t) &\leq -\lambda \sum_{i=1}^N c_i |\rho_i(t)|^2 + d \sum_{i=1}^N c_i |e_i(t) + \tilde{e}_i|^2 \\ &\leq -\lambda \sum_{i=1}^N c_i |\rho_i(t)|^2 + d(1 + \varepsilon) \sum_{i=1}^N c_i |\tilde{e}_i|^2 + \frac{1+\varepsilon}{\varepsilon} d \sum_{i=1}^N c_i |e_i(t)|^2 \\ &\leq -\lambda \sum_{i=1}^N c_i |\rho_i(t)|^2 + \frac{2d\delta_1(1+\varepsilon)}{1-2\delta_1} \sum_{i=1}^N c_i |\rho_i(\Delta_t)|^2 + \frac{1+\varepsilon}{\varepsilon} d \sum_{i=1}^N c_i |e_i(t)|^2 \\ &\leq -\gamma_1 \sum_{i=1}^N c_i |\rho_i(t)|^2. \end{aligned} \quad (14)$$

Moreover, we know that

$$V(t) = \sum_{i=1}^N c_i |\rho_i(t)|^2 \geq \min_{i \in \mathcal{N}^*} \{c_i\} |\rho(t)|^2. \quad (15)$$

Therefore, we finally get

$$|\rho(t)|^2 \leq \frac{1}{\min_{i \in \mathcal{N}^*} \{c_i\}} V(0) \exp\{-\gamma_1 t\}.$$

Thus, according to Definition 1, we can conclude that system (1) and system (2) can reach exponential synchronization. The proof is completed.

Remark 3. In Theorem 1, we select τ_1 satisfying the vital condition $\tau_1 < \tau_1^*$, where τ_1^* is the unique solution to the equation

$$-(\lambda - \gamma_1)G_2(\tau_1) + \frac{1 + \varepsilon}{\varepsilon}dG_1(\tau_1) + \frac{2d\delta_1(1 + \varepsilon)}{1 - 2\delta_1}$$

The following analysis can concisely prove the uniqueness of the solution. Denote $f(\tau_1) = -(\lambda - \gamma_1)G_2(\tau_1) + \frac{1 + \varepsilon}{\varepsilon}dG_1(\tau_1) + \frac{2d\delta_1(1 + \varepsilon)}{1 - 2\delta_1}$. Then, it can be deduced that $f(0) = -(\lambda - \gamma_1) + \frac{2d\delta_1(1 + \varepsilon)}{1 - 2\delta_1} < 0$, $f'(\tau_1) > 0$ and $f(+\infty) = +\infty$. Thus, based on the continuity and monotonicity of function f , we can obtain the unique positive solution τ_1^* such that $f(\tau_1^*) = 0$.

Next, we consider another event-triggered mechanism to naturally exclude the Zeno phenomenon, which can be regarded as a special case of PETC. The event-triggered control with dwell-time (ETCDT) is defined as follows

$$t_{k+1} = \inf \left\{ t > t_k + \tau_1 \mid \sum_{i=1}^N c_i |\rho_i(t) - \rho_i(t_k)|^2 \geq \delta_2 \sum_{i=1}^N c_i |\rho_i(t_k)|^2 \right\}. \quad (16)$$

Corollary 1. System (1) and system (2) can reach exponential synchronization under ETCDT (16) if Assumption 1 and the following conditions are satisfied

$$0 < \epsilon < 1, \lambda > \gamma_1, \delta_2 \in \left(0, \frac{\epsilon(\lambda - \gamma_1)(1 - \epsilon)}{(1 - \epsilon)d + \epsilon(\lambda - \gamma_1)} \right),$$

and $\tau_1 \in (0, \bar{\tau}_1)$, where $\bar{\tau}_1$ is the unique solution to the equation $-(\lambda - \gamma_1)G_2(\tau_1) + dG_1(\tau_1) = 0$.

Proof. Similar to the proof of Theorem 1, we also need to discuss the two cases and prove the inequality (14) is satisfied. If $t \in [t_k, t_k + \tau_1]$, it follows from Case I of Theorem 1 that (14) can be obtained, where in $G_1(\tau_1)$, $\frac{2d^2}{1 - 2\delta_1} \rightarrow d^2$, and in $G_2(\tau_1)$, $\frac{2d}{\varepsilon_3(1 - 2\delta_1)} \rightarrow \frac{d}{\varepsilon_3}$. If $t \in [t_k + \tau_1, t_{k+1})$, based on Assumption 1, we can get

$$\begin{aligned} \dot{V}(t) &\leq -\lambda \sum_{i=1}^N c_i |\rho_i(t)|^2 + d \sum_{i=1}^N c_i |\rho_i(t) - \rho_i(\xi_t)|^2 \\ &\leq \left(-\lambda + \frac{(1 - \epsilon)\delta_2 d}{\epsilon(1 - \epsilon - \delta_2)} \right) \sum_{i=1}^N c_i |\rho_i(t)|^2 \\ &\leq -\gamma_1 \sum_{i=1}^N c_i |\rho_i(t)|^2, \end{aligned}$$

The proof is completed.

4. Alternate periodic event-triggered control

In this section, we will propose alternate periodic event-triggered control, which combines the advantages of ETC and AIC. This enables the new ETC to achieve flexible intermittent ETC while naturally excluding the Zeno phenomenon. Both APETC and alternate event-triggered control with dwell-time (AETCDT) are considered.

The APETC $u_k(t)$ is designed as follows

$$u_k(t) = \begin{cases} -d_i \rho_i(t_k), & t \in [t_k, s_k), \\ 0, & t \in [s_k, t_{k+1}), \end{cases}$$

where $[t_k, s_k)$ represents the k -th control interval, $[s_k, t_{k+1})$ signifies the k -th rest interval, $s_k - t_k$ and $t_{k+1} - s_k$ denote the control width and the rest width respectively, in which $k \in \mathbb{Z}$. The APETC is determined by two alternate event-triggered conditions, that is, the control instant t_{k+1} and the rest instant s_k are respectively determined by

$$\begin{aligned} s_k &= \inf \left\{ m\tau_1 > t_k \mid \sum_{i=1}^N c_i |\rho_i(m\tau_1) - \rho_i(t_k)|^2 \geq \delta_1 \sum_{i=1}^N c_i |\rho_i(t_k)|^2 \right\}, \\ t_{k+1} &= \inf \left\{ \theta\tau_2 > s_k \mid |\sigma(t)|^2 \geq \frac{\alpha}{(t - t_0)^2} \sum_{i=1}^N c_i |\rho_i(s_k)|^2 + \delta_3 \sum_{i=1}^N c_i |\rho_i(\theta\tau_2)|^2 e^{-\beta(t - s_k)} \right\}, \end{aligned} \quad (17)$$

where $\delta_1, \delta_3, \alpha$ and β are positive constants and $|\sigma(t)|^2$ will be defined later.

Remark 4. A novel APETC is designed, in which the control input generator and event generator will be in the working state when the work condition is satisfied and will have rest intervals when the rest condition occurs. Compared with Reference [41], the IC considered is in the periodic case, while the IC in this paper is in the aperiodic case. In contrast to the traditional AIC [33], the control and rest intervals are determined by the state of the system instead of being predesigned. Reference [37] investigated the dynamic properties of NNs by adopting intermittent control based on time-triggered control. The state of the system is sampled on every fixed time interval in this literature, which may bring about limitations. APETC can lower the unnecessary updates of control signals and the waste of resources by introducing the periodic event-triggered mechanism.

Firstly, we give a lemma to obtain a lower bound on the control interval and an upper bound on the rest interval.

Lemma 2. The control interval and the rest interval are estimated as follows

$$T_1 = s_k - t_k \geq T_1^*, T_2 = t_{k+1} - s_k \in (0, T_2^*),$$

where $T_1^* = \frac{1}{\Lambda_1} \ln \left(\frac{\delta_1 \Lambda_1}{\Lambda_2} + 1 \right)$ and T_2^* is the unique solution to the equation $-\varphi_1(T_2^*) + \varphi_2(T_2^*) = 0$, in which

$$\begin{aligned} \varepsilon_5 = \max_{i \in \mathcal{V}} \{ \varepsilon_{i5} \}, \varepsilon_{i5} = -b_i + 2\rho_{i1} + 2\Gamma^* M \sum_{j=1}^N r_{ij}, \Lambda_1 = \max_{i \in \mathcal{V}} \{ \Lambda_{i1} \}, \Lambda_{i1} = -b_i + 1 + 2\rho_{i1}^2 + 3\Gamma^* M \sum_{j=1}^N a_{ij} + d_i, \Lambda_2 = \\ \max_{i \in \mathcal{V}} \{ \Lambda_{i2} \}, \Lambda_{i2} = b_i + 2\rho_{i1}^2 + 2\Gamma^* M \sum_{j=1}^N a_{ij} + d_i, \beta > \varepsilon_5. \text{ Also, we define } |\sigma(t)|^2 = \frac{\varepsilon_4^*}{\varepsilon_4} (\exp\{\varepsilon_4(t - s_k)\} - 1) \sum_{i=1}^N c_i |\rho_i(s_k)|^2, \text{ in which} \\ \varepsilon_4 = \max_{i \in \mathcal{V}} \{ \varepsilon_{i4} \}, \varepsilon_{i4} = -b_i + 1 + 2\rho_{i1}^2 + 3\Gamma^* M \sum_{j=1}^N a_{ij}, \varepsilon_4^* = \max_{i \in \mathcal{V}} \{ \varepsilon_{i4}^* \}, \varepsilon_{i4}^* = b_i + 2\rho_{i1}^2 + 2\Gamma^* M \sum_{j=1}^N a_{ij}. \end{aligned}$$

Proof. Assuming that $\eta_i(t) = \rho_i(t) - \rho_i(\xi_t)$, we can obtain

$$\begin{aligned} & \frac{d \left(\sum_{i=1}^N c_i |\eta_i(t)|^2 \right)}{dt} \\ & \leq \sum_{i=1}^N c_i 2\eta_i^T(t) \left(-b_i \rho_i(t) + f_i(y_i(t), t) - f_i(x_i(t), t) + \sum_{l=1}^M r_l \sum_{j=1}^N a_{ij}^{(l)} \rho_j(t) - d_i \rho_i(\xi_t) \right) \\ & \leq \sum_{i=1}^N c_i \left(-b_i + 1 + 2\rho_{i1}^2 + 3\Gamma^* M \sum_{j=1}^N a_{ij} + d_i \right) |\eta_i(t)|^2 + \sum_{i=1}^N c_i \left(b_i + 2\rho_{i1}^2 + 2\Gamma^* M \sum_{j=1}^N a_{ij} + d_i \right) |\rho_i(\xi_t)|^2 \\ & \leq \Lambda_1 \sum_{i=1}^N c_i |\eta_i(t)|^2 + \Lambda_2 \sum_{i=1}^N c_i |\rho_i(\xi_t)|^2. \end{aligned}$$

Thus, we can derive that when $t \in [t_k, s_k)$,

$$\sum_{i=1}^N c_i |\eta_i(t)|^2 \leq \frac{\Lambda_2}{\Lambda_1} (e^{\Lambda_1(t-t_k)} - 1) \sum_{i=1}^N c_i |\rho_i(\xi_t)|^2.$$

Since $s_k = \inf \left\{ m\tau_1 > \xi_t \mid \sum_{i=1}^N c_i |\rho_i(m\tau_1) - \rho_i(\xi_t)|^2 \geq \delta_1 \sum_{i=1}^N c_i |\rho_i(\xi_t)|^2 \right\}$, this fact yields that the next trigger time s_k must satisfy that

$$\delta_1 \sum_{i=1}^N c_i |\rho_i(\xi_t)|^2 \leq \frac{\Lambda_2}{\Lambda_1} (e^{\Lambda_1(s_k-t_k)} - 1) \sum_{i=1}^N c_i |\rho_i(\xi_t)|^2.$$

Therefore, we have

$$s_k - t_k \geq \frac{1}{\Lambda_1} \ln \left(\frac{\delta_1 \Lambda_1}{\Lambda_2} + 1 \right).$$

When $t \in [s_k, t_{k+1})$, let $\eta_i^*(t) = \rho_i(t) - \rho_i(s_k)$. Similarly, we can obtain

$$\begin{aligned} & \frac{d \left(\sum_{i=1}^N c_i |\eta_i^*(t)|^2 \right)}{dt} \\ & \leq \sum_{i=1}^N c_i (-b_i + 1 + 2\rho_{i1}^2 + 3\Gamma^* M \sum_{j=1}^N a_{ij}) |\eta_i^*(t)|^2 + \sum_{i=1}^N c_i (b_i + 2\rho_{i1}^2 + 2\Gamma^* M \sum_{j=1}^N a_{ij}) |\rho_i(s_k)|^2 \\ & \leq \varepsilon_4 \sum_{i=1}^N c_i |\eta_i^*(t)|^2 + \varepsilon_4^* \sum_{i=1}^N c_i |\rho_i(s_k)|^2. \end{aligned}$$

Hence, it can be derived that

$$\sum_{i=1}^N c_i |\eta_i^*(t)|^2 \leq \frac{\varepsilon_4^*}{\varepsilon_4} (e^{\varepsilon_4(t-s_k)} - 1) \sum_{i=1}^N c_i |\rho_i(s_k)|^2 \triangleq |\sigma(t)|^2.$$

According to the event-triggered mechanism (17), we consider the following equation

$$\begin{aligned} & \frac{\varepsilon_4^*}{\varepsilon_4} (e^{\varepsilon_4(t-s_k)} - 1) \sum_{i=1}^N c_i |\rho_i(s_k)|^2 \\ &= \frac{\alpha}{(t-t_0)^2} \sum_{i=1}^N c_i |\rho_i(s_k)|^2 + \delta_3 \sum_{i=1}^N c_i |\rho_i(\theta\tau_2)|^2 e^{-\beta(t-s_k)}. \end{aligned} \quad (18)$$

When $t \in [s_k, t_{k+1})$, one has

$$\begin{aligned} \dot{V}(t) &= \sum_{i=1}^N c_i 2\rho_i^T(t) \left(-b_i \rho_i(t) + f_i(y_i(t), t) - f_i(x_i(t), t) + \sum_{l=1}^M r_l \sum_{j=1}^N a_{ij}^{(l)} \rho_j(t) \right) \\ &\leq \sum_{i=1}^N c_i \left(-2b_i + 2\rho_{i1} + 2\Gamma^* M \sum_{j=1}^N r_{ij} \right) |\rho_i(t)|^2 \\ &\leq \varepsilon_5 V(t). \end{aligned} \quad (19)$$

Therefore, we can get

$$\sum_{i=1}^N c_i |\rho_i(t)|^2 \leq \sum_{i=1}^N c_i |\rho_i(s_k)|^2 e^{\varepsilon_5(t-s_k)}. \quad (20)$$

Combining (18) and (20), we consider the following equation

$$\frac{\varepsilon_4^*}{\varepsilon_4} (e^{\varepsilon_4(t-s_k)} - 1) = \frac{\alpha}{(t-t_0)^2} + \delta_3 e^{-(\beta-\varepsilon_5)(t-s_k)}.$$

Let $\varphi_1(t-s_k) = \frac{\varepsilon_4^*}{\varepsilon_4} (e^{\varepsilon_4(t-s_k)} - 1)$ and $\varphi_2(t-s_k) = \frac{\alpha}{(t-t_0)^2} + \delta_3 e^{-(\beta-\varepsilon_5)(t-s_k)}$. Then, it can be derived that

$$\begin{cases} \varphi_1(0) = 0, \\ \dot{\varphi}_1(t-s_k) > 0, \quad \forall t \geq s_k \\ \varphi_1(t-s_k) \rightarrow \infty, \quad t-s_k \rightarrow \infty; \end{cases}$$

and

$$\begin{cases} \varphi_2(0) = \frac{\alpha}{(s_k-t_0)^2} + \delta_3 > 0, \\ \dot{\varphi}_2(t-s_k) \leq 0, \quad \forall t \geq s_k. \end{cases}$$

Thus, we can conclude that the above equation $\varphi_1(t-s_k) = \varphi_2(t-s_k)$ has a unique solution on $(0, +\infty)$, which implies the rest interval has an upper bound T_2^* . The proof is completed. Then, based on Lemma 2, the main results with regard to the exponential synchronization of system (1) and system (2) with APETC are given.

Theorem 2. System (1) and system (2) can reach exponential synchronization under APETC (17) if Assumption 1, Theorem 1 and the following condition are satisfied

$$\varepsilon_5 \left(\frac{\lceil \frac{T_2^*}{\tau_2} \rceil}{\tau_2} \right) \leq \frac{\gamma_1}{3} \left(\frac{\lceil \frac{T_1^*}{\tau_1} \rceil}{\tau_1} \right). \quad (21)$$

Proof. Define Lyapunov function $V(t) = \sum_{i=1}^N c_i |\rho_i(t)|^2$. According to (14) and (19), we can obtain

$$\dot{V}(t) = \begin{cases} -\gamma_1 V(t), & t \in [t_k, s_k), \\ \varepsilon_5 V(t), & t \in [s_k, t_{k+1}). \end{cases}$$

When $t \in [0, s_0)$,

$$V(t) \leq V(0) e^{-\gamma_1 t}.$$

When $t \in [s_0, t_1)$,

$$V(t) \leq V(s_0)e^{\varepsilon_5(t-s_0)} \leq V(0)e^{-\gamma_1 s_0 + \varepsilon_5(t-s_0)}.$$

When $t \in [t_1, s_1)$,

$$V(t) \leq V(t_1)e^{-\gamma_1(t-t_1)} \leq V(0)e^{-\gamma_1 s_0 + \varepsilon_5(t-s_0) - \gamma_1(t-t_1)}.$$

Based on the condition (21), we can calculate that

$$\begin{aligned} & \varepsilon_5(t_{k+1} - s_k) - \gamma_1(s_k - t_k) \\ & \leq -\frac{2\gamma_1}{3}(s_k - t_k) \\ & \leq -\varepsilon_5(t_{k+1} - s_k) - \gamma_1(s_k - t_k) \\ & \leq -\varepsilon_6(t_{k+1} - t_k), \end{aligned} \tag{22}$$

where $\varepsilon_6 = \min\{\varepsilon_5, \gamma_1\}$.

When $t \in [t_k, s_k)$, according to (22), it can be deduced that

$$\begin{aligned} V(t) & \leq V(t_k)e^{-\gamma_1(t-t_k)} \\ & \leq V(0)e^{-\gamma_1(t-t_k) + \sum_{h=1}^k \varepsilon_5(t_h - s_{h-1}) - \sum_{k=1}^k \gamma_1(s_{k-1} - t_{k-1})} \\ & \leq V(0)e^{-\varepsilon_6(t-t_0)}. \end{aligned}$$

When $t \in [s_k, t_{k+1})$,

$$\begin{aligned} V(t) & \leq V(s_k)e^{\varepsilon_5(t-s_k)} \\ & \leq V(0)e^{\varepsilon_5(t-s_k) - \sum_{h=1}^{k+1} \gamma_1(s_{h-1} - t_{h-1}) + \sum_{h=1}^k \varepsilon_5(t_h - s_{h-1})} \\ & \leq V(0)e^{\varepsilon_5(t-s_k) - \gamma_1(s_k - t_k) - \varepsilon_6(t_k - t_0)} \\ & \leq V(0)e^{-\varepsilon_6(t-t_0)}. \end{aligned}$$

Therefore, according to (15) and Definition 1, we can conclude that system (1) and system (2) can reach exponential synchronization under APETC (17).

Consider alternate event-triggered control with dwell-time (AETCDT) as follows

$$\begin{aligned} s_k & = \inf \left\{ t > t_k + \tau_1 \mid \sum_{i=1}^N c_i |\rho_i(t) - \rho_i(t_k)|^2 \geq \delta_2 \sum_{i=1}^N c_i |\rho_i(t_k)|^2 \right\}, \\ t_{k+1} & = \inf \left\{ t > s_k + \tau_2 \mid |\sigma(t)|^2 \geq \frac{\alpha}{(t-t_0)^2} \sum_{i=1}^N c_i |\rho_i(s_k)|^2 + \delta_3 \sum_{i=1}^N c_i |\rho_i(t)|^2 e^{-\beta(t-s_k)} \right\}. \end{aligned} \tag{23}$$

Corollary 2. System (1) and system (2) can reach exponential synchronization under AETCDT (23) if Assumptions 1, Corollary 1 and the following condition are satisfied

$$\varepsilon_5(T_2^* \vee \tau_2) \leq \frac{\gamma_1}{3}(T_1^* \vee \tau_1).$$

Proof. Based on the analysis above, the proof is similar to the proof of Theorem 2. Hence, we omit it here.

Remark 5. In comparison with the traditional ETC applied in neural networks in previous work, which requires continuous-time state observations to monitor the event-triggered condition, we employ PETC and APETC in Theorem 1 and Theorem 2 respectively. They can avoid continuous-time state observations while naturally eliminating the Zeno phenomenon, which is capable of saving the transmitted resources. Additionally, we adopt ETCDT and AETCDT in Corollary 1 and Corollary 2 respectively. It uses the time-regularization design frameworks to guarantee a positive minimum inter-event time, which can also exclude the Zeno phenomenon. Thus, the classes of event-triggered mechanisms in this paper outperform the existing event-triggered mechanisms.

According to the obtained results, in order to illustrate how to select suitable and feasible parameters to achieve exponential synchronization under APETC, the selection process is given below.

- **Step1** Initialize the parameters of the system (1), including b_i and $a_{ij}^{(l)}$.
- **Step2** By choosing $\delta_1, \delta_3, \alpha, \beta, \sigma$ and d_i , calculate ε_5 and γ_1 .
- **Step3** Solve Eq. (10) in Theorem 1 and Eq. (19) in Lemma 2 to obtain τ_1 and τ_2 , respectively.
- **Step4** Substitute in data to check whether conditions (21) in Theorem 2 hold. If it does, output the values of τ_1 and τ_2 . Otherwise, return to Step 2.

5. Numerical simulations

In this section, several numerical simulations are presented to demonstrate the validity of our theoretical results. Let $M = 3, N = 15$. Then the coupling matrices of MMNs containing 15 nodes are established on the three subdigraphs $(\mathcal{G}, (a_{ij}^{(1)})_{15 \times 15})$, $(\mathcal{G}, (a_{ij}^{(2)})_{15 \times 15})$ and $(\mathcal{G}, (a_{ij}^{(3)})_{15 \times 15})$ revealed in the left part of Fig. (1). Subsequently, we synthesize the three subdigraphs above, which can obtain a composite digraph $(\mathcal{G}, (a_{ij})_{15 \times 15})$ shown in the right part of Fig. 1.

Then the non-zero elements of three matrices $(a_{ij}^{(1)})_{15 \times 15}$, $(a_{ij}^{(2)})_{15 \times 15}$ and $(a_{ij}^{(3)})_{15 \times 15}$ are provided as follows.

- $a_{1,2}^{(1)} = a_{3,1}^{(1)} = a_{4,10}^{(1)} = a_{9,3}^{(1)} = a_{8,13}^{(2)} = a_{11,10}^{(2)} = a_{12,14}^{(2)} = a_{13,15}^{(2)} = a_{14,12}^{(2)} = a_{15,9}^{(2)} = a_{2,3}^{(3)} = a_{5,4}^{(3)} = a_{6,7}^{(3)} = 0.001$,
- $a_{1,4}^{(1)} = a_{6,12}^{(1)} = a_{7,1}^{(1)} = a_{11,5}^{(1)} = a_{9,10}^{(2)} = a_{9,15}^{(2)} = a_{10,9}^{(2)} = a_{10,11}^{(2)} = a_{12,13}^{(2)} = a_{13,8}^{(2)} = a_{13,12}^{(2)} = a_{13,14}^{(2)} = a_{2,7}^{(3)} = a_{3,4}^{(3)} = a_{4,3}^{(3)} = a_{4,5}^{(3)} = a_{5,6}^{(3)} = a_{7,6}^{(3)} = 0.002$,
- $a_{3,2}^{(3)} = a_{7,2}^{(3)} = 0.003, a_{8,15}^{(2)} = a_{14,11}^{(2)} = a_{15,8}^{(2)} = 0.007$,
- $a_{1,6}^{(1)} = a_{2,8}^{(1)} = a_{5,1}^{(1)} = a_{13,7}^{(2)} = a_{11,14}^{(2)} = a_{6,5}^{(3)} = 0.005$,

Apparently, subdigraphs $(\mathcal{G}, (a_{ij}^{(1)})_{15 \times 15})$, $(\mathcal{G}, (a_{ij}^{(2)})_{15 \times 15})$ and $(\mathcal{G}, (a_{ij}^{(3)})_{15 \times 15})$ are not strongly connected, while the digraph $(\mathcal{G}, (a_{ij})_{15 \times 15})$ is strongly connected, where $a_{ij} = \max\{a_{ij}^{(1)}, a_{ij}^{(2)}, a_{ij}^{(3)}\}$.

Choose $r_l = 1, b_i = 0.75, d_i = 2, \tau_1 = 0.0013, \tau_2 = 0.0014, \delta_1 = 0.1, \delta_2 = 0.02, \delta_3 = 5, \varepsilon = 0.01, \epsilon = 0.5, \gamma_1 = 1.45, \alpha = 0.001, \beta = 35, i \in \mathcal{N}, l \in \mathcal{M}$. Meanwhile, set the activation function $f_i(x) = 0.76 \tanh(x)$. According to the expression of $f_i(x)$, we take $\rho_{11} = 0.76, i \in \mathcal{N}$, which implies Assumption 1 is satisfied. Based on the above data, we can calculate that $\Gamma^* = 1, \lambda = 1.998$. Then, through some simple calculation, we can determine that the unique solutions to equations are $\tau_1^* = 0.0018, T_1^* = 0.0241$ and $T_2^* = 0.1042$ separately. Consequently, it is obvious that all the conditions in Theorems 1, 2 and Corollary 1, 2 are satisfied. For arbitrary chosen initial values, the simulation results are exhibited in Figs. 2–5. Fig. 2 shows the trajectories of the same drive system (1) under PETC, ETCDT, APETC and AETCDT. Fig. 3 presents the trajectories of response system (2). Fig. 4 reflects the trajectories of control system (4). Fig. 6 shows the event-triggered instants. The trajectories of error system (3) eventually approaching zero are provided in Fig. 5, which illustrate the validity of our results.

Comparison 1. To compare the PETC, ETCDT, APETC and AETCDT, Fig. 6 is exhibited. The number of event triggers is 112 under PETC and 134 under ETCDT, while 78 event triggers are adopted under APETC and 86 event triggers are adopted under AETCDT. It can be clearly seen that the event-triggered instants under APETC and AETCDT are considerably smaller than that under PETC and ETCDT. Also, the number of event triggers under AETCDT overtakes that under APETC. Therefore, the APETC we employ is superior to PETC, ETCDT and AETCDT in terms of transmitted information and control cost.

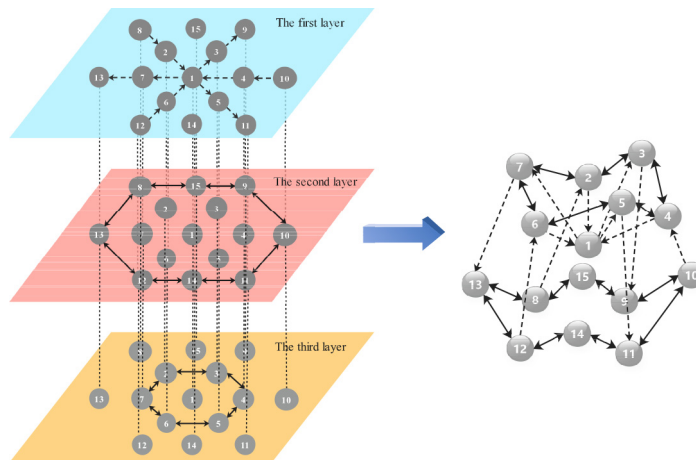


Fig. 1. The digraph $\mathcal{G}$ with 15 nodes: in the left one, the first layer is subdigraph $(\mathcal{G}, (a_{ij}^{(1)})_{15 \times 15})$, the second layer is subdigraph $(\mathcal{G}, (a_{ij}^{(2)})_{15 \times 15})$, the third layer is subdigraph $(\mathcal{G}, (a_{ij}^{(3)})_{15 \times 15})$ and the right one is composite digraph $(\mathcal{G}, (a_{ij})_{15 \times 15})$.

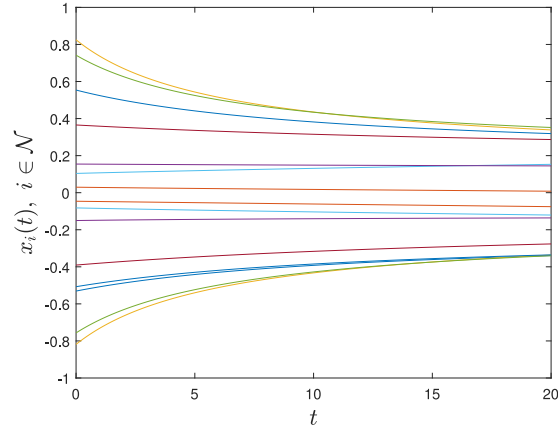
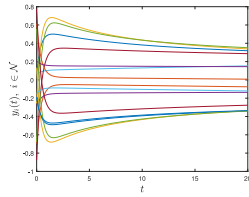
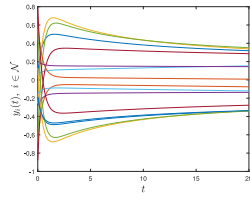


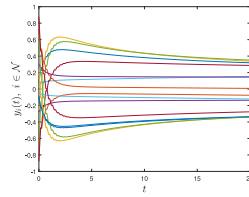
Fig. 2. The trajectories of drive system (1).



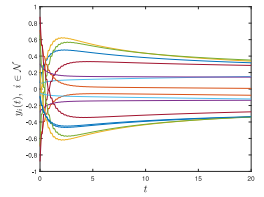
(a) The response system (2) under PETC.



(b) The response system (2) under ETCDT.

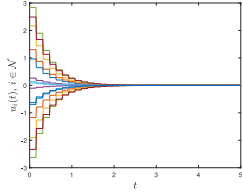


(c) The response system (2) under APETC.

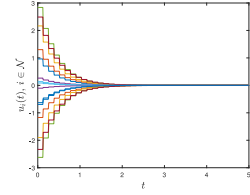


(d) The response system (2) under AETCDT.

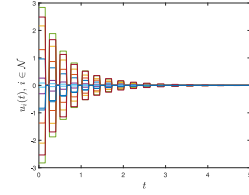
Fig. 3. The trajectories of response system (2).



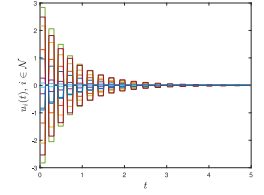
(a) The control system (4) under PETC.



(b) The control system (4) under ETCDT.

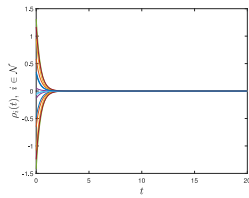


(c) The control system (4) under APETC.

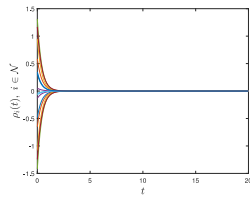


(d) The control system (4) under AETCDT.

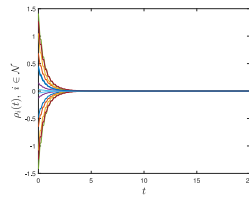
Fig. 4. The trajectories of control system (4).



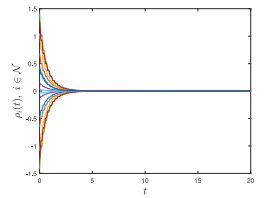
(a) The error system (3) under PETC.



(b) The error system (3) under ETCDT.



(c) The error system (3) under APETC.



(d) The error system (3) under AETCDT.

Fig. 5. The trajectories of error system (3).

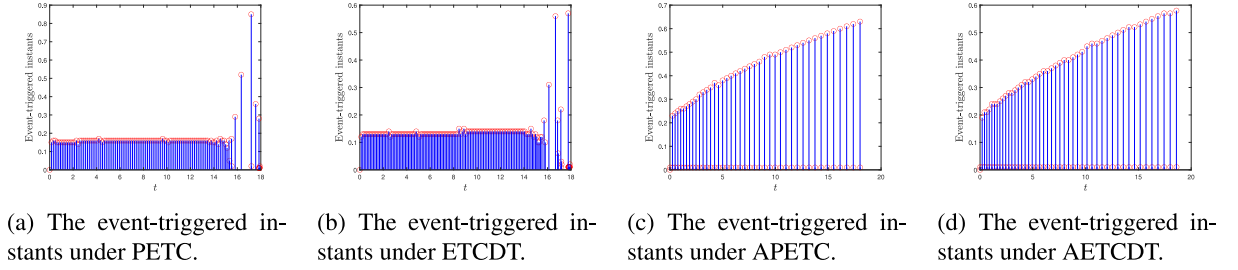


Fig. 6. The event-triggered instants.

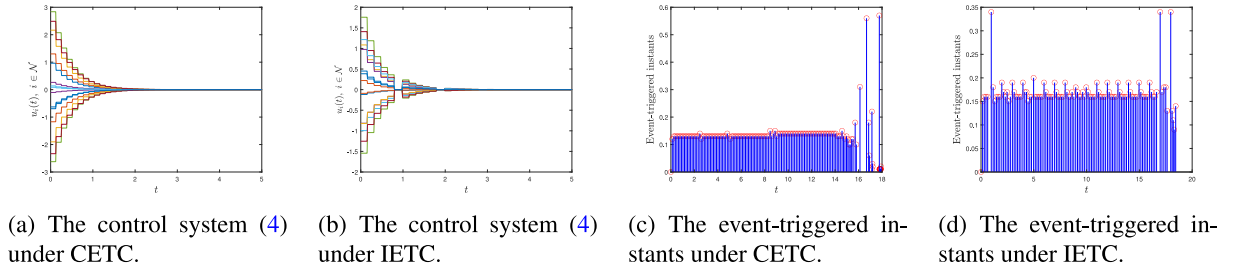


Fig. 7. The comparison between CETC and IETC on the control system and event-triggered instants.

Table 1

The number of event triggers between CETC, IETC and APETC

Control strategy	The number of event triggers
CETC	143
IETC	110
APETC	78

Comparison 2. In order to compare APETC, CETC of [48], and intermittent event-triggered control (IETC) of [12], Fig. 7 and Table 1 are given. Fig. 7 shows the trajectories of control input and the number of event triggers under CETC and IETC. By analyzing the statistics, we can derive that the sampling frequency of the MNNs under APETC is decreased by 29.09% compared with IETC and the sampling frequency of the MNNs under APETC is reduced by approximately 45.45% compared with CETC, which indicates that APETC can dramatically reduce the number of event triggers.

6. Conclusion

In this paper, we proposed a novel APETC to research the exponential synchronization of MNNs. Before deriving our theoretical results, two indispensable lemmas are established. By means of Lyapunov method and graph theory, synchronization criteria for MNNs under PETC, ETCDT, APETC and AETCDT were derived. Then, some numerical simulations demonstrated the validity of our synchronization criteria for MNNs. It is suggested that the proposed APETC could significantly decrease the number of event triggers and naturally avoid the Zeno phenomenon as it integrates AIC and PETC, which outperforms either the conventional AIC or ETC in the existing literature. Considering the existence of the delay, a novel APETC will be designed to study the exponential synchronization of delay MNNs in the future. It is worth pointing out that every node needs to be controlled under PETC, ETCDT, AETCDT and APETC in this paper, and thus the conditions may be complex, which limits the scope of their utility in practice. In the future, we will devote to seeking other control methods to simplify conditions, such as pinning APETC. Besides, we note that the control gain in this paper is constant and cannot be adjusted according to systems' evolution results. In the future, we will consider combining adaptive control with the APETC to further study the dynamical behaviors of delay MNNs.

CRedit authorship contribution statement

Dongsheng Xu: Conceptualization, Methodology, Investigation, Writing - original draft. **Chennuo Dai:** Writing - review & editing. **Huan Su:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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